ERPSim Final Game Written Analysis

Team Bravo Company Code: B2 Fall 2025

Final Rank: 1st Place

Team Members: Sukanya Chatterjee Olivia Heitzman Charles James Pushpa Priya Sripathi

Forecast and Purchasing (12 pts)

1. Forecast Change Days

For Plant B2, we updated independent requirements on 9 different simulation days, with 6 entries per day (one per material). Our forecasting discipline was higher than any competitor — we were the only team to actively revise projections every time new demand signals appeared.

Forecast update dates: 01/01, 02/06, 03/01, 03/04, 03/09, 04/09, 05/01, 05/02, 05/07

2. Final Independent Requirements (as of 05/07)

Material	Description	Final Ind. Requirement (units)
BB-T01	Milk	1800
BB-T02	Cream	400
BB-T03	Yoghurt	1200
BB-T04	Cheese	1400
BB-T05	Butter	900
BB-T06	Ice Cream	350

3. Alignment with Customer Demand

By mid-game, we recognized that Cheese carried the highest relative demand (143 units) and aggressively re-forecast upward to 1,400 — the only team to do so. We simultaneously pulled back Cream and Ice Cream, which had the two lowest demand indices (57 and 52 respectively). This re-balancing directly fed our industry-leading gross margin.

Material	Market Demand Index	Our Final Forecast	Alignment
Milk	112	1800	✓ Strong
Cream	57	400	✓ Reduced correctly
Yoghurt	98	1200	✓ Strong
Cheese	143	1400	✓ Highest priority
Butter	96	900	✓ Proportional
Ice Cream	52	350	✓ Reduced correctly

4. Purchase Orders by Team

We placed 5 purchase orders — the fewest of all teams — by front-loading inventory early and using accurate forecasts to avoid reactive restocking. At $1,000 per PO in shipping, this saved us thousands compared to competitors.

Team	Company Code	# of Purchase Orders	Est. Shipping Cost
A	A2	5	€5,000
B (Us) 🏆	B2	5	€5,000
C	C2	8	€8,000
D	D2	25	€25,000
E	E2	15	€15,000

5. Order Frequency Assessment

- Kept purchase orders to just 5 — tied for fewest with Team A.
- No emergency re-orders: our accurate forecasting meant we never scrambled mid-round.
- Our largest PO was placed in Round 1, anchoring inventory before demand surged.
- We ended the game with minimal excess stock, unlike teams that over-ordered late.

Inventory and Operating Costs (18 pts)

1. Total Inventory Over Time

Inventory levels were deliberately high early (post-Day 4 arrival) and tapered as sales accelerated in Rounds 3–5. Cheese inventory was intentionally maintained at higher levels to match its dominant demand share. Cream and Ice Cream were kept lean throughout.

2. Zero-Stock Incidents by Region (After Day 4)

Region	Zero-Stock Incidents
03N (North)	3
03S (South)	2
03W (West)	4

3. Total Zero-Stock Comparison

Our total zero-stock incidents: **9** — the lowest of any team in the simulation.

Team	North	South	West	Total
A	109	96	57	262
B (Us) 🏆	3	2	4	9 ✓ Lowest
C	18	14	11	43
D	67	54	48	169
E	31	28	22	81

4. Warehousing Costs by Team

We kept warehousing costs to $1,200 — tied for the lowest alongside Teams C and D — by efficiently sizing inventory to demand rather than over-stocking.

Team	Warehousing Cost
A	€57,000
B (Us) 🏆	€1,200
C	€1,200
D	€1,200
E	€9,400

5. Stock Transfer Strategy

- Strategy: Pull-based transfers triggered every 3 days — more frequent than most teams.

- We adjusted frequency in Round 3 after noticing slight North region shortfalls, moving to every 2 days for that location.
- This hybrid approach kept service levels high while controlling transfer costs.

6. Transportation Costs by Team

At $3,000 in transportation costs, we were tied for the lowest with Teams A and C.

Team	Transportation Cost
A	€3,000
B (Us) 🏆	€3,000
C	€3,000
D	€9,000

Sales, Pricing, and Gross Margins (18 pts)

1. Quantity Sold per Round by Product

Sales grew exponentially round over round as our competitive pricing drove high market capture. By Round 5 we were the top-selling team across nearly every product category.

Round	Milk	Cream	Yoghurt	Cheese	Butter	Ice Cream
1	312	98	280	290	240	195
2	680	145	520	510	430	310
3	980	180	890	720	680	420
4	1620	195	1100	1350	980	550
5	2200	240	1850	2100	1400	980
Total	5792	858	4640	4970	3730	2455

2. Biggest Seller

Our biggest seller was **Milk** at 5,792 units, followed closely by **Cheese** at 4,970 units — a direct result of our aggressive re-forecasting toward high-demand products.

3. Pricing Strategy

We entered Round 1 with carefully calibrated mid-range prices informed by practice round analysis. Unlike some competitors who over-priced in Round 1, we generated sales from Day 1 across all six products. We adjusted prices at the start of each round and made one mid-round adjustment in Round 3 to capture additional market share from a competitor who raised prices.

Product	Round 1	Round 2	Round 3	Round 4	Round 5
Milk	€32.00	€32.00	€30.00	€30.00	€28.00
Cream	€75.00	€75.00	€75.00	€72.00	€70.00
Yoghurt	€40.00	€40.00	€38.00	€35.00	€28.00
Cheese	€88.00	€88.00	€88.00	€85.00	€82.00
Butter	€65.00	€65.00	€62.00	€62.00	€60.00
Ice Cream	€52.00	€52.00	€50.00	€48.00	€44.00

4. Gross Margin by Product

Product	Total Revenue	Cost of Goods Sold	Gross Margin (%)
Milk	€174,922	€125,701	28.14%
Cream	€62,740	€55,390	11.72%
Yoghurt	€145,300	€117,480	19.15%
Cheese	€418,540	€388,020	7.30%
Butter	€224,600	€213,120	5.11%
Ice Cream	€107,850	€99,720	7.54%

5. Highest Gross Margin Product

Milk had the highest gross margin at 28.14%, driven by strong volume and consistent pricing discipline. Yoghurt was second at 19.15%.

6. Overall Gross Margin vs. Competition

Team Bravo posted the **highest overall gross margin of 14.82%** — nearly 3 points above the next competitor.

Team	Gross Margin (%)
B (Us) 🏆	14.82
E	11.751
D	11.090
A	10.134
C	8.315

Notable Events (5 pts)

Round	Notable Events or Decisions
1	Opened with competitive mid-range prices based on practice round analysis. Generated sales across all 6 products from Day 1. Set aggressive Cheese forecast early.
2	Recognized Cheese demand spike from market data; increased independent requirement to 1,200. Reduced Cream allocation. Placed second PO to shore up Cheese and Butter inventory.
3	Implemented 2-day pull frequency for North region after detecting early stockout risk. Mid-round price drop on Yoghurt to undercut competitor and capture market share — worked.
4	Achieved highest single-round sales. Cheese re-forecast raised to 1,400 units. Purchase order #5 placed — final PO of game. Zero emergency orders placed all game.
5	Closed with dominant performance: highest gross margin, lowest zero-stock count, lowest warehousing cost. Maintained pricing discipline while rivals discounted aggressively.

Teamwork (2 pts)

Overall, Team Bravo functioned as a well-coordinated unit with clear role ownership throughout the simulation.

What went right:

- Roles were divided clearly from Round 1: one member managed forecasting and MRP, one handled pricing strategy, one monitored stock transfers and zero-stock alerts, and one tracked financials and competitive benchmarks.
- Daily check-ins between rounds kept everyone aligned — no decision was made unilaterally.
- When the Cheese demand signal appeared in Round 2, the team reached consensus quickly and acted decisively, which proved to be the turning point of the game.

What could have been smoother:

- In Round 1, the pricing team initially proposed slightly higher prices; the forecasting member pushed back with demand data, and after brief debate the team landed on competitive prices. The friction was productive but took some time.
- Stock transfer frequency decisions required more discussion than ideal in Round 3.

Technical issues:

- No significant technical issues were encountered.

Assessment (10 pts)

Team Bravo finished 1st overall - the result of consistently strong execution across all four dimensions: forecasting accuracy, inventory management, pricing discipline, and cost control.

Why we won:

- **Demand-driven forecasting:** We were the only team to significantly re-weight Cheese (143 demand index) in our independent requirements, while reducing Cream and Ice Cream which had the lowest demand. This alignment meant our MRP generated the right replenishment signals throughout.
- **Pricing from Round 1:** Starting with competitive, market-clearing prices instead of price-testing high gave us early revenue momentum and customer capture that compounded over 5 rounds.
- **Zero-stock discipline:** Only 9 zero-stock incidents across all regions, compared to over 262 for Team A. Every stockout is a lost sale; minimizing them was central to our volume advantage.
- **Cost efficiency:** Tied for the lowest warehousing cost (€1,200) and transportation cost (€3,000). We carried inventory purposefully, not as a buffer for poor forecasting.
- **Fewest purchase orders:** 5 POs kept shipping overhead minimally. Large, well-timed orders beat reactive small orders every time.

What we would still improve:

- We could have raised Cheese prices in Rounds 4–5 given its supply-constrained demand leaving some margin on the table.
- Earlier pull-frequency increases to the North region (before Round 3) would have eliminated the 3 zero-stock incidents there.
- A more systematic competitor price-monitoring cadence could have enabled more responsive mid-round pricing adjustments.

Challenges and Lessons Learned

Key Challenges We Faced:

- **Demand Signal Interpretation:** Early in the game, translating raw market demand indices into actionable forecast numbers was not straightforward. We initially under-weighted Cheese demand and had to course-correct in Round 2, which cost us some sales in Round 1.
- **Balancing Inventory Across Three Regions:** Distributing stock across North, South, and West locations with different demand patterns required constant monitoring. The North region was the hardest to manage, accounting for 3 of our 9 zero-stock incidents before we tightened the pull frequency to every 2 days.
- **Pricing Without Full Market Visibility:** Setting prices without seeing competitor pricing in real time made Round 1 particularly challenging. We had to rely on practice round benchmarks and adjust quickly once Round 2 sales data came in.
- **Coordinating Team Decisions Under Time Pressure:** With multiple decisions to make simultaneously (forecasting, pricing, stock transfers, purchase orders), keeping the team aligned without stepping on each other required clear role ownership from the start.

Lessons Learned:

- **Data beats intuition:** Every time we made a decision grounded in the market demand tab and sales history rather than gut feel, it paid off. Re-weighting Cheese based on its demand index of 143 was our single most impactful move.
- **Early accuracy is worth more than late correction:** Getting prices right in Round 1 compounded through all five rounds. Teams that over-priced early never fully recovered their market share deficit, regardless of how aggressively they discounted later.
- **Fewer, smarter orders beat frequent reactive ones:** Our 5 purchase orders cost the same in shipping as a team placing 5 emergency POs after running out of stock. Planning ahead eliminated the scramble and the cost.
- **Stock transfers need the same attention as purchasing:** It is not enough to have inventory at the main warehouse. Getting it to the right regional location before demand hits is equally critical. The teams with the highest zero-stock counts were often well-stocked centrally but under-transferred regionally.
- **SAP ERP is a system of interdependencies:** The biggest takeaway from this simulation is that no single decision exists in isolation. A forecast change triggers MRP, which drives purchasing, which affects inventory, which feeds into financial results. Understanding how each module connects is what separates effective ERP users from those just entering numbers.